PRODUCT & TECHNICAL REQUIREMENTS DOCUMENT

Omni-Channel Appointment Booking Agent

Document: Sample PRD / Technical Requirements Sheet
Version: 0.1 — Draft for technical discovery
Status: Proposed; validate with stakeholders
Audience: Product, Engineering, AI/ML, Integrations, QA, Security and Operations

Purpose

Define a reusable AI appointment-booking agent that understands customer requests, checks live availability, books, reschedules or cancels appointments, and confirms outcomes across messaging and voice. It should support spas, salons, clinics, doctors' offices and other appointment-based businesses through configurable business profiles—not separate custom agents for every industry.

Core principle: One shared booking orchestration layer and business configuration model, with channel-specific experiences for text and voice.

Channels in scope

Channel	Experience	Interaction
Facebook Messenger	Text chatbot	Customer messages and agent replies in Messenger
Instagram Messenger	Text chatbot	Customer messages and agent replies in Instagram DMs
Website embedded chat	Text chatbot widget	Conversational booking embedded on business website
Phone	Voice agent	Speech-based conversation; agent listens, responds and books

1. Product Overview

The agent acts as a virtual receptionist. It identifies intent, understands the requested service, collects required details, checks the business's actual availability and completes the action only after the customer confirms.

1.1 Goals

- Enable 24/7 appointment booking without requiring staff intervention.
- Provide consistent booking capability across social messaging, website chat and phone.
- Reduce missed inquiries, manual scheduling effort and booking errors.
- Support multiple locations, services, staff, durations, business hours and booking policies.
- Allow business configuration without custom development for each new tenant.

1.2 Suggested initial exclusions

- Clinical diagnosis, treatment recommendations or medical advice beyond approved safety scripts.
- Full CRM replacement, payroll, inventory management or marketing automation.
- Complex recurring/group bookings unless explicitly added to scope.
- Payments/deposits unless payment integration and compliance requirements are approved.

2. Users & Roles

Role	Needs / permissions
Customer / caller	Discover services; ask questions; book, reschedule or cancel; receive confirmation.
Business admin	Configure profile, locations, services, hours, policies, integrations and agent behavior.
Receptionist / staff	View/manage appointments, handle escalations and review permitted conversation context.
Platform admin	Manage tenants, channel connections, permissions, monitoring and platform controls.

3. Core Functional Requirements

Priority: **P0** launch-critical; **P1** important; **P2** future/optional.

ID	Requirement	Priority	Description
FR-001	Business identification & configuration	P0	Resolve tenant from channel connection, website embed or inbound phone number. Load business services, locations, hours, policies, language and rules.
FR-002	Intent recognition	P0	Recognize booking, availability, reschedule, cancellation, FAQs, human assistance and unsupported requests.
FR-003	Service selection	P0	Clarify service; support categories, aliases, duration, price/range, add-ons, eligibility and intake fields.
FR-004	Location selection	P0	Identify preferred location for multi-location businesses; never mix calendars across locations.
FR-005	Staff/provider preference	P1	Allow specific provider or 'any available'; respect service qualification rules.
FR-006	Live availability	P0	Query scheduler using duration, hours, staff, location, buffers, time zone and booking constraints.
FR-007	Slot suggestions	P0	Offer clear available options; understand natural-language dates/times and clarify ambiguity.
FR-008	Pre-booking review	P0	Summarize service, location, date/time, provider, duration, fees and customer details; obtain explicit confirmation.
FR-009	Create appointment	P0	Create in source-of-truth scheduler; prevent duplicates and recheck conflicts.
FR-010	Customer details	P0	Collect only required details; validate and confirm spelling/digits when needed.
FR-011	Reschedule	P0	Find/verify booking, check new availability, summarize change, confirm and update scheduler.

ID	Requirement	Priority	Description
FR-012	Cancel	P0	Find/verify booking, explain configured policy, confirm cancellation and update scheduler.
FR-013	Confirmation/reminders	P0	Provide booking reference/details; support configured reminders through integrated channels.
FR-014	Business FAQs	P0	Answer from approved content: services, prices, hours, address, accessibility, preparation and policies.
FR-015	Human handoff	P0	Escalate on request, low confidence, exceptions, complaints or failure; pass summary and collected details.
FR-016	Conversation continuity	P1	Maintain session context; retrieve cross-channel context only when identity and consent permit.
FR-017	Languages	P1	Support configurable languages; voice requires compatible speech recognition and synthesis.
FR-018	Business rules	P0	Apply lead time, booking horizon, cancellation window, buffers, eligibility, deposits and required fields.
FR-019	Failure recovery	P0	Handle unclear input, no slots, timeouts and interruptions; never claim success without scheduler confirmation.
FR-020	Analytics/audit	P0	Record channel, intent, funnel stage, outcome, handoff, failure reason and latency subject to privacy rules.

4. Channel-Specific Requirements

4.1 Messaging: Facebook Messenger, Instagram Messenger and website chat

- Support inbound/outbound text and channel-appropriate formatting; show typing/processing indicators where supported.
- Use quick replies, buttons or structured choices where available, with text fallback.
- Handle multi-part messages, typos, corrections, short replies and natural-language date/time expressions.
- Website widget supports configurable branding, greeting, launcher and responsive layout.
- Respect Meta platform permissions, messaging windows, opt-in rules and approved outbound templates.
- Provide a concise text equivalent if a channel cannot render a requested format.

4.2 Phone voice agent

- Answer inbound calls with configured greeting; identify business/location from number or routing metadata.
- Use speech recognition, dialogue orchestration and text-to-speech with barge-in and natural turn-taking.
- Speak clearly, avoid long lists, repeat important details and confirm critical fields when confidence is low.
- Handle silence, noise, interruptions, DTMF where available, voicemail and dropped calls.
- Offer live transfer or callback capture when staff are unavailable.
- Provide automated-agent and recording/consent disclosures where required.
- Never invent availability or claim a booking is complete before scheduler confirmation.

4.3 Channel parity

All channels use the same booking rules and source-of-truth scheduling APIs. Presentation may differ, but eligibility, availability, confirmation, cancellation and audit outcomes must remain consistent.

5. Business Configuration & Scheduling Model

Area	Fields / behavior
Business profile	Name, category, time zone, locale, contact details, locations, channel IDs, brand voice and escalation contacts.
Services	Name, category, description, duration, price/range, buffers, add-ons, eligibility and intake fields.
Availability	Hours, holidays, staff schedules, rooms/equipment, capacity, lead time and booking horizon.
Policies	Cancellation/reschedule, deposits, late arrival, no-show, age requirements and customer-facing wording.
Providers	Staff identity, qualified services, locations, schedule, breaks, preference rules and calendar connection.
Knowledge base	Approved FAQs, directions, preparation instructions, accessibility, pricing disclaimers and business answers.

6. Integrations & Architecture

Layer	Responsibility
Channel adapters	Meta webhooks/APIs, website chat SDK, telephony/SIP or voice provider; normalize inbound events.
Conversation orchestration	Session state, intent/entity extraction, dialogue policy, tool selection, business context and channel rendering.
Booking service/API	Canonical operations: availability search, create, retrieve, reschedule, cancel and confirm.
Integration connectors	Adapters for scheduling/CRM systems; map external IDs, calendars, staff, services and statuses.
Tenant configuration	Business profiles, secrets, channel mapping, feature flags, policies, knowledge and permissions.
Observability/governance	Metrics, traces, audit events, quality evaluation, access control, retention and incident monitoring.

6.1 Integration expectations

- Documented, versioned APIs with authentication, authorization, idempotency keys and consistent errors.

- Webhooks/events for external appointment changes where supported.
- Business scheduling platform remains the source of truth; avoid conflicting shadow calendars.
- Retries with backoff for transient failures and reconciliation for uncertain outcomes.
- Encrypt and tenant-scope channel credentials and provider secrets.
- Create an integration certification checklist for every supported scheduler/telephony provider.

7. Data, Privacy & Security

- Tenant isolation across conversations, customer records, calendars and credentials.
- Collect only data necessary for the request; define retention and deletion policies.
- Encrypt data in transit and at rest; role-based access and audit logs.
- Protect against prompt injection, malicious links, data exfiltration and unauthorized tool/API calls.
- For healthcare, assess applicable legal/regulatory/contractual obligations; minimize sensitive health data and do not diagnose or prescribe.
- Support consent/disclosure requirements for calls, recordings, messaging and data use.
- Redact logs where configured; do not store unnecessary sensitive or payment-card data.

8. Non-Functional Requirements

Area	Sample target / requirement
Availability	Proposed 99.9% monthly availability for core orchestration, excluding documented third-party outages; confirm SLO.
Latency	Text p95 under 3 seconds for non-scheduler turns; measure scheduler latency separately. Voice pauses should be minimized; set pilot target.
Scalability	Support multi-tenant concurrent sessions; load-test peak traffic and provider limits.
Reliability	Idempotent operations, duplicate prevention, timeout handling and reconciliation.
Accessibility	Website widget supports keyboard navigation, screen readers, contrast and responsive layouts.
Maintainability	Modular adapters, versioned APIs, configuration-driven rules and regression tests.
Monitoring	Dashboards/alerts for errors, conversion, latency, provider health, handoffs and channel failures.

9. Reference Conversation & Booking Flow

1. **Welcome:** greet customer and identify business using channel context.
2. **Intent:** determine booking, reschedule, cancellation or question.
3. **Service:** identify requested service and clarify ambiguity.
4. **Location/provider:** collect preferences if applicable.
5. **Time:** ask for date/time or range; clarify ambiguous expressions.
6. **Availability:** query live scheduler and offer valid slots.
7. **Customer details:** collect and validate required information.
8. **Review:** summarize appointment and policies; obtain explicit confirmation.
9. **Commit:** create appointment and verify success response.
10. **Confirm:** provide reference, date/time, location, service and next steps.

Required exception paths

- No availability: ask for alternate date/time/provider/location.
- Slot conflict: re-query and offer alternatives.
- Customer correction: update state and re-check affected fields.
- Unclear speech/text: ask one concise clarification; do not guess critical details.
- Integration unavailable: explain limitation and offer handoff/callback; do not represent an unconfirmed request as booked.
- Medical/urgent request: follow approved safety routing; no diagnosis or treatment advice.

10. Acceptance Criteria / Definition of Done

- New booking completes end-to-end on all four channels.
- Agent checks live availability and commits to configured source-of-truth scheduler.
- Explicit confirmation required before create, reschedule or cancel.
- Retries and duplicate events do not create duplicate appointments.
- Slot conflict triggers fresh availability check and alternatives.
- Reschedule/cancel verifies booking and applies configured policies.
- Supported business configuration changes do not require code deployment.
- Human handoff includes summary, collected details and escalation reason.
- Analytics/logs capture outcomes while honoring privacy and retention settings.
- Tests cover happy path, ambiguity, no availability, API failure, interruption and recovery.

11. Analytics & Success Metrics

Metric	Definition	Breakdown
Booking conversion	Completed bookings ÷ eligible booking conversations	Channel, business, service, location
Completion rate	Successful scheduler commits ÷ booking attempts	Failure reason, scheduler
Self-service resolution	Conversations completed without human handoff	Intent, channel, business type
Handoff rate	Transferred/callback conversations ÷ total	Reason, channel, time
Booking error rate	Incorrect, duplicate or unconfirmed bookings ÷ completed bookings	Error type, integration
Customer satisfaction	Post-interaction rating or survey; define method	Channel, flow, business
Response latency	P50/P95 first response and tool completion	Channel, integration, period

12. Suggested Delivery Phases

Phase	Scope	Exit criteria
0 — Discovery	Confirm scheduler(s), channel/provider constraints, data model, pilot businesses, privacy and ownership.	Approved requirements, architecture, integration plan and test scenarios.
1 — Booking MVP	Website chat + one messaging channel + one scheduler; booking, availability, FAQs and handoff.	Pilot meets agreed quality, reliability and booking accuracy targets.
2 — Channel expansion	Add remaining Meta channel, voice, reschedule/cancel, reminders and analytics.	Channel parity, voice quality and monitoring validated.
3 — Platform scale	Multiple scheduler connectors, self-serve onboarding, multilingual support and advanced rules.	Repeatable onboarding/support across business categories.

13. Open Questions for Technical Discovery

- Which scheduling systems are required first (industry software, Google/Microsoft calendars, clinic systems, custom APIs)?
- One scheduler per tenant initially, or multiple connected calendars?
- Which telephony provider, number provisioning and call-transfer model?
- Initial languages, countries, time zones and business categories?
- Are deposits/payments, insurance details or patient intake forms included? What compliance review is needed?
- What identity verification is required before modifying an existing booking?
- Expected traffic, concurrency, uptime SLOs and response targets?
- Who owns onboarding, knowledge updates, escalation coverage and incident response?

14. Sample User Stories

Story	User story
Customer booking	As a customer, I want to book by chat or phone so I can secure an appointment outside business hours.
Availability	As a customer, I want actual available times so I can choose a slot that fits my schedule.
Reschedule	As a customer, I want to change my appointment when my plans change.
Business admin	As an admin, I want to configure services, durations, hours and policies so bookings follow business rules.
Staff handoff	As a receptionist, I want a conversation summary so I can help without asking the customer to repeat everything.

End of sample document. Validate priorities, service-level targets, integrations and compliance requirements with stakeholders before implementation.